

IntelliCrash Explainable AI (XAI) Forensic Report

Incident Severity: **MANEUVER (HARD CORNERING) DETECTED**

AI Decision Reasoning:

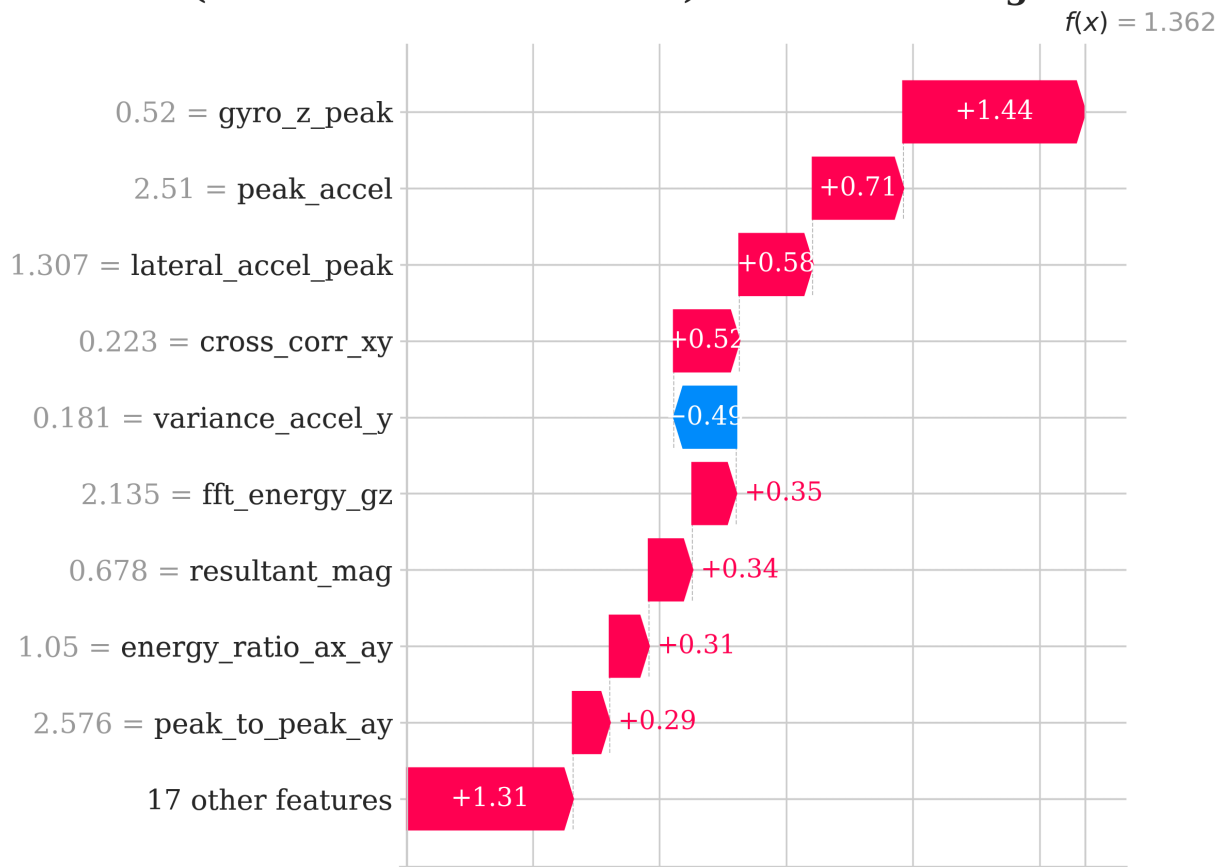
The XGBoost model classified this event based on the following primary physical factors:

- gyro_z_peak was measured at 0.52, which increased the raw hazard score by 1.44 points.
- peak_accel was measured at 2.51, which increased the raw hazard score by 0.71 points.
- lateral_accel_peak was measured at 1.31, which increased the raw hazard score by 0.58 points.

Understanding the Visual Chart Below:

The chart visualizes the mathematical breakdown of the AI's decision. The gray $E[f(X)]$ at the bottom is the baseline average score. The red bars push the hazard score higher (more dangerous), while the blue bars push the score lower (safer). The final $f(x)$ at the top is the absolute hazard rating assigned to this event.

XAI (Best Auto-TOPSIS Model): Hard Cornering Incident Reasoning



Conclusion: This forensic report is generated autonomously by the SHAP TreeExplainer to provide transparency for emergency dispatchers and insurance evaluators.